2. Analysis of Existing Systems

2.1 Technical Specifications of Existing Tea Harvesting Machines

Table 01: Technical Specifications of Common Tea Harvesting Machines Used in Sri Lanka and Similar Plantation Systems

Machine	Power Source	Weight/ kg	Cutting Mechanism	Output / Efficiency	Figures
Kawasaki NV60H	Petrol (TU26 Mitsubishi engine, 2 Stroke, Air cool)	9.4 (Engine 6.2kg)	Shear blades (Blade length 600mm)	0.7–1.0 acre/day, ~15× faster than manual plucking	
Single Hand Electric Tea Harvester	Battery powered	1.8	Shear blades (Blade length 330mm)	30+ kg/hr (similar speed)	
Kawasaki Impeller machine	Battery powered	3.5	Shear blades with impeller/fan	Faster than manual, lower than NV60H	
Static Teeth Machine	Battery powered	1.75	Fixed teeth cutting system	Very low efficiency	
TRI selective harvesting shear	Manual	<1	Scissor mechanism	4.3 kg/hour, ~2× faster than hand plucking	

2.2 Performance Comparison of Tea Harvesting Machines

Table 02: Performance Evaluation of Tea Harvesting Machines Based on Tea Quality and Operational Limitations

Machine	Tea Quality Performance	Major Specifications	Main Limitations
Kawasaki NV60H	Low quality due to broken leaves, coarse leaves, and stem inclusion	High cutting speed, widely used	No selective plucking → harvests bud + coarse leaves together; high leaf damage; operator fatigue; post-sorting required
Single Hand Electric Tea Harvester	Moderate quality (non-selective)	Lightweight, portable	No bud-selectivity → mixed leaf harvest; battery dependency; limited coverage width; operator wrist fatigue during long use
Kawasaki Impeller machine	Moderate–low quality	Lightweight, improved handling	Leaf bruising due to impeller action; poor selectivity
Static Teeth Machine	Very poor quality	Simple mechanical design	Jamming in dense tea bushes; uneven cutting; high coarse leaf content; unsuitable for Sri Lanka terrain
TRI selective harvestin g shear	High quality (closest to hand plucking)	Simple, low cost	Low productivity; fatigue; not suitable for steep slopes

2.3 Comparative Rating of Tea Harvesting Machines

Table 03: Comparative Performance Rating of Tea Harvesting Machines Based on Key Evaluation Criteria

Machine	Harvesting Efficiency	Tea Quality	Selectivity	Ergonomics (Operator Comfort)	Overall Rating (Average)
Kawasaki NV60H	5	2	1	1	2.25
Single Hand Electric Harvester	4	2	1	3	2.5
Kawasaki Impeller Machine	3	3	2	4	3
Static Teeth Machine	3	3	2	3	2.75
TRI selective harvesting shear	2	5	4	4	3.75

Scale

- 5 = Excellent
- 4 = Good
- 3 = Moderate
- 2 = Poor
- 1 = Very Poor